

CHE 201 Fall 2019 HW 7

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TOTAL POINTS

90.5 / 100

QUESTION 1

11 15 / 15

✓ - **0 pts** Correct

- **1 pts** a) Wrong schematic (A2 should be $< A1$)
- **1 pts** a) wrong or missing system boundary
- **1.5 pts** c) Missing mass balance
- **4 pts** d) Wrong value, wrong calculation or missing

T2. Correct answer is $h_2 = 99.5$ Btu/lb and $T_2 = -2.4$ °F

- **2 pts** e) State 1 is superheated refrigerant
- **2 pts** e) State 2 is saturated refrigerant (L+V mixture)
- **4 pts** e) Wrong or missing P-v diagram

QUESTION 2

2 2 13 / 15

- **0 pts** Correct
- **1 pts** a) Wrong or missing schematic
- **1 pts** a) wrong or missing system boundary
- **2 pts** b) Wrong or missing eng model
- **1.5 pts** c) Missing mass balance
- **2 pts** d) Missing or wrong exit area. $A_2 = 0.06$ m²
- **4 pts** d) mass flow rate = 4.5 kg and $A_2 = 0.06$ m²
- ✓ - **2 pts** e) $V_2 > V_1$ (expansion)
- **4 pts** a) Missing or wrong T-v diagram

QUESTION 3

3 3 13 / 15

- **0 pts** Correct
- **1 pts** a) Wrong schematic
- **1 pts** a) Missing or wrong system boundary
- **2 pts** b) State 1 is water vapor
- **4 pts** b) Wrong or missing P-v diagram
- ✓ - **2 pts** b) $V_2 > V_1$ (expansion)
- **2 pts** b) State 2 is saturated vapor

- **2 pts** c) Missing eng model

- **3 pts** d) Missing simplified mass rate and energy rate balance.

- **1.5 pts** d) Missing simplified mass rate

- **4 pts** e) $W = 3581$ kW

QUESTION 4

4 4 20 / 20

✓ - **0 pts** Correct

- **3 pts** T-V figure
- **1 pts** T-V figure, $v_1 = ?$ $v_2 = ?$
- **5 pts** Q calculation incorrect
- **3 pts** scheme incorrect, you mention $\Delta PE = 0$ but your input and output is not on the same level
- **1 pts** T-V figure where is step 1 and 2
- **3 pts** scheme incorrect. input? output?
- **5 pts** m incorrect
- **5 pts** scheme

QUESTION 5

5 5 12 / 15

- **0 pts** Correct

✓ - **3 pts** scheme, storage tank that is 15 m above reservoir

- **3 pts** storage tank that is 15 m above reservoir; the pump and the storage tank should be on the same level

- **5 pts** w calculation incorrect

- **3 pts** unit in KW

- **3 pts** $h_1 = ?$ $h_2 = ?$

- **3 pts** engineering model

QUESTION 6

6 6 17.5 / 20

- **0 pts** Correct

- **1 pts** a) did not label the system boundary

- **1 pts** a) wrong schematic
- **4 pts** b) missing the engineering model
- **2 pts** c) missing the mass rate balance
- **2 pts** d) missing or wrong energy rate balance
- **4 pts** d) used wrong enthalpy values to solve for x
- **5 pts** d) $x = 0.194$
- **2.5 pts** e) State 1 is compressed liquid
- **5 pts** e) wrong or missing sketch of the process on

a T-v diagram

✓ - **2.5 pts** e) $v_2 > v_1$

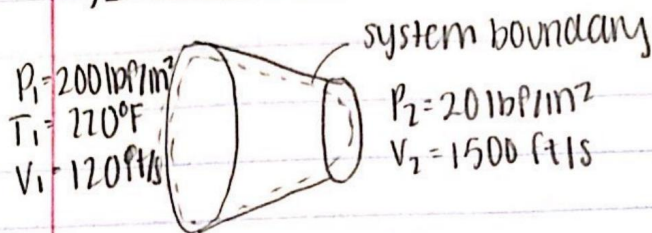
- **2.5 pts** e) missing point 2 on the T-v diagram
- **2.5 pts** e) $T_1 > T_2$

CHE 201
homework 7

1.

Refrigerant 134A enters a well insulated nozzle at 200 lbf/in², 220°F with a velocity of 120 ft/s and exits at 20 lbf/in² with a velocity of 1500 ft/s. For steady state & neglecting PE effects determine exit temp

a) schematic



$T_{sat} @ 200 \text{ lbf/in}^2 = 125.3$
 $220 > 125.3 \rightarrow \text{sat vap}$

b) engineering model

- control volume
- steady state
- $\Delta PE = 0$
- insulated well, $\dot{Q}_{cv} = 0$
- $\dot{W}_{cv} = 0$; work flow only (mass enters & exits cv)
- one inlet, one outlet

c) mass rate balance:

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e \Rightarrow \boxed{\dot{m}_i = \dot{m}_e = \dot{m}}$$

energy rate balance

$$\frac{dm_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) + g(z_1 - z_2) \right]$$

$$\dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) \right] = 0$$

$$\boxed{0 = (h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right)}$$

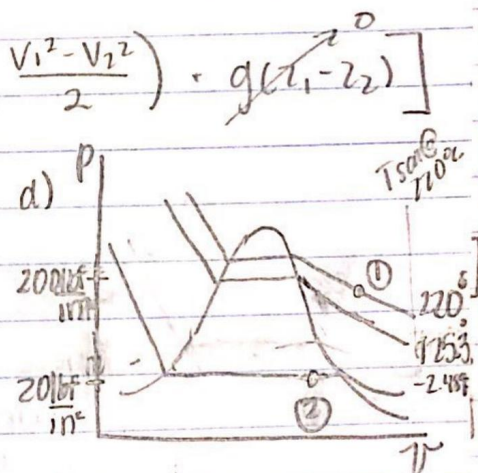
d) $h_1 = 144.15 \text{ Btu/lb}$ (A12E)

$$0 = \left(144.15 \frac{\text{Btu}}{\text{lb}} - h_2 \right) + \left(\frac{(120 \frac{\text{ft}}{\text{s}})^2 - (1500 \frac{\text{ft}}{\text{s}})^2}{2} \right) \cdot \frac{\text{lbf}}{32.174 \frac{\text{lbm} \cdot \text{ft}}{\text{s}^2}} \cdot \frac{1 \text{ Btu}}{778.17 \text{ ft} \cdot \text{lbf}}$$

$$0 = 144.15 \frac{\text{Btu}}{\text{lb}} - h_2 - 44.64 \frac{\text{Btu}}{\text{lb}}$$

$$h_2 = 99.5 \frac{\text{Btu}}{\text{lb}} \rightarrow \text{in vapor dome at } 20 \text{ lbf/in}^2$$

$$\hookrightarrow \boxed{T_f = -2.48^\circ \text{F}}$$



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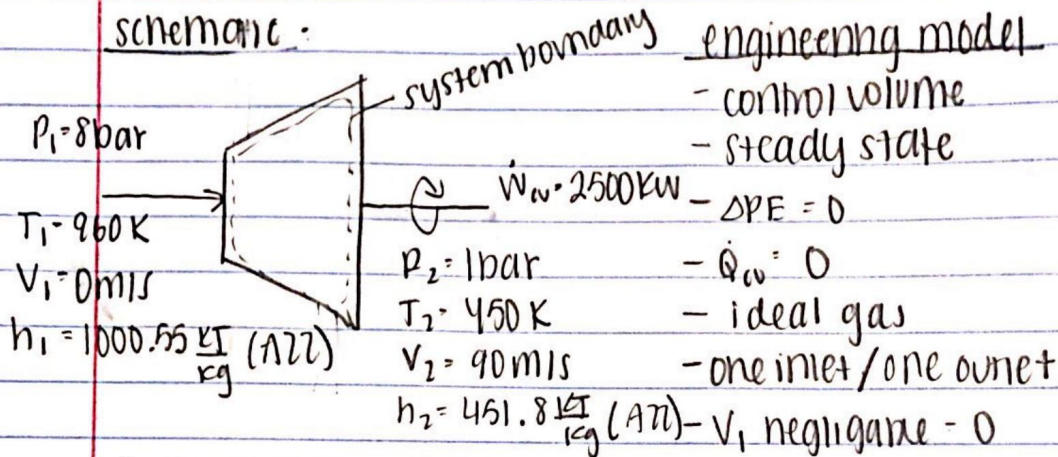
✓ - 0 pts Correct

- 1 pts a) Wrong schematic (A2 should be $< A1$)
- 1 pts a) wrong or missing system boundary
- 1.5 pts c) Missing mass balance
- 4 pts d) Wrong value, wrong calculation or missing T2. Correct answer is $h_2 = 99.5 \text{ Btu/lb}$ and $T_2 = -2.4 \text{ oF}$
- 2 pts e) State 1 is superheated refrigerant
- 2 pts e) State 2 is saturated refrigerant (L+V mixture)
- 4 pts e) Wrong or missing P-v diagram

2.

Air expands through a turbine from 8 bar, 960 K to 1 bar 450 K. The inlet velocity is small compared to exit velocity of 90 m/s. The turbine operates at steady state and develops a power output of 2500 kW. Heat transfer b/w the turbine & surroundings is negligible & PE is negligible. Air is ideal gas. Find mass flow rate & exit area.

schema :



$$1 \text{ bar} = 10^5 \text{ Nm}^2 / 10^3 \text{ mm}^2 = 100 \text{ kPa}$$

$$R = 8.314 \text{ kJ} / \text{kmol K} \cdot \frac{1 \text{ kmol}}{28.97 \text{ kg}} = 0.2869$$

c) mass rate balance

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e \Rightarrow \dot{m}_i = \dot{m}_e = \dot{m}$$

energy rate balance

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) + g(z_1 - z_2) \right]$$

$$\dot{W}_{cv} = \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) \right]$$

$$d) 2500 \text{ kW} = \dot{m} \left[(1000.55 \frac{\text{kJ}}{\text{kg}} - 451.8 \frac{\text{kJ}}{\text{kg}}) + \left(\frac{(0 \frac{\text{m}}{\text{s}})^2 - (90 \frac{\text{m}}{\text{s}})^2}{2} \cdot \frac{1 \text{ N}}{1 \text{ kg m/s}^2} \cdot \frac{1 \text{ kJ}}{10^3 \text{ Nm}} \right) \right]$$

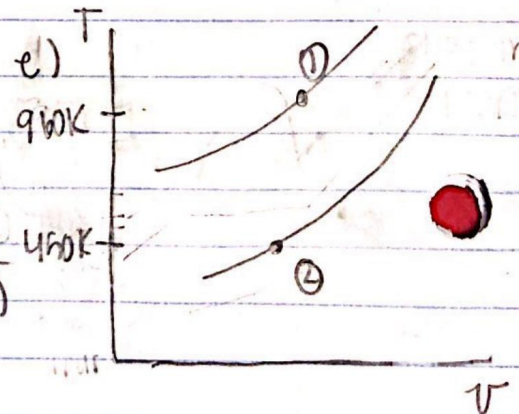
$$2500 \text{ kW} = \dot{m} \left[(548.75 \frac{\text{kJ}}{\text{kg}} + (-4.05 \frac{\text{kJ}}{\text{kg}})) \right]$$

$$\dot{m} = 4.59 \frac{\text{kg}}{\text{s}}$$

$$\dot{m} = \frac{A_2 V_2}{v} \quad p v = R T \Rightarrow v = \frac{R T}{p}$$

$$\dot{m} = \frac{A_2 V_2 p_2}{R T_2} \Rightarrow 4.59 \frac{\text{kg}}{\text{s}} = \frac{A_2 (90 \frac{\text{m}}{\text{s}}) (100 \text{ kPa})}{(0.2869) (450 \text{ K})}$$

$$A_2 = 0.0658 \text{ m}^2$$



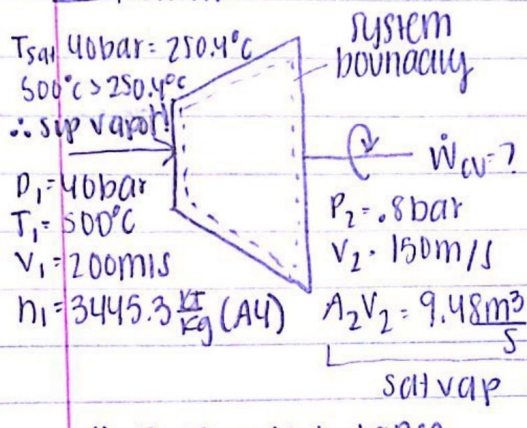
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- 0 pts Correct
- 1 pts a) Wrong or missing schematic
- 1 pts a) wrong or missing system boundary
- 2 pts b) Wrong or missing eng model
- 1.5 pts c) Missing mass balance
- 2 pts d) Missing or wrong exit area. $A_2 = 0.06 \text{ m}^2$
- 4 pts d) mass flow rate = 4.5 kg and $A_2 = 0.06 \text{ m}^2$
- ✓ - 2 pts e) $V_2 > V_1$ (expansion)
- 4 pts a) Missing or wrong T-v diagram

3.

Water vapor enters a turbine operating at steady state at 500°C , 40 bar, w/ velocity of 200 m/s, and expands adiabatically to exit, where it is saturated vapor at .8 bar, w/ velocity of 150 m/s & volumetric flow rate of $9.48 \text{ m}^3/\text{s}$. Neglecting PE, determine power in kW.

Schematic



engineering model

- control volume
- steady state
- adiabatic expansion ($\dot{Q}_{cv} = 0$)
- sch vapor at exit
- $\Delta PE = 0$
- one inlet / one outlet

d) mass rate balance

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e \Rightarrow \boxed{\dot{m}_i = \dot{m}_e = \dot{m}}$$

energy rate balance:

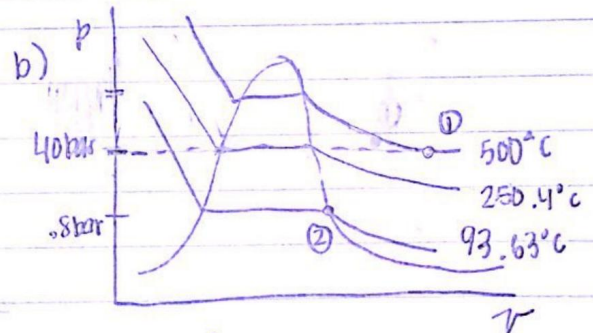
$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) + g(z_1 - z_2) \right]$$

$$\boxed{\dot{W}_{cv} = \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) \right]}$$

e) $\dot{m} = \frac{A_2 V_2}{v_2}$

$\rightarrow$ sch vap @ .8 bar
 $\Rightarrow 2.087 \frac{\text{m}^3}{\text{kg}}$ (A3)

$$\dot{m} = \frac{9.48 \frac{\text{m}^3}{\text{s}}}{2.087 \frac{\text{m}^3}{\text{kg}}} = 4.54 \frac{\text{kg}}{\text{s}}$$



$$\dot{W}_{cv} = 4.54 \frac{\text{kg}}{\text{s}} \left[\left(3445.3 \frac{\text{kJ}}{\text{kg}} - 2665.8 \frac{\text{kJ}}{\text{kg}} \right) + \left(\frac{(200 \frac{\text{m}}{\text{s}})^2 - (150 \frac{\text{m}}{\text{s}})^2}{2} \right) \cdot \frac{1 \text{ W}}{1 \text{ kg} \cdot \frac{1}{10^3 \text{ N} \cdot \text{m}}} \right]$$

$$\dot{W}_{cv} = 4.54 \frac{\text{kg}}{\text{s}} \left[788.25 \frac{\text{kJ}}{\text{kg}} \right] \quad 1 \frac{\text{kJ}}{\text{s}} = 1 \text{ kW}$$

$$\boxed{\dot{W}_{cv} = 3578.65 \text{ kW}}$$

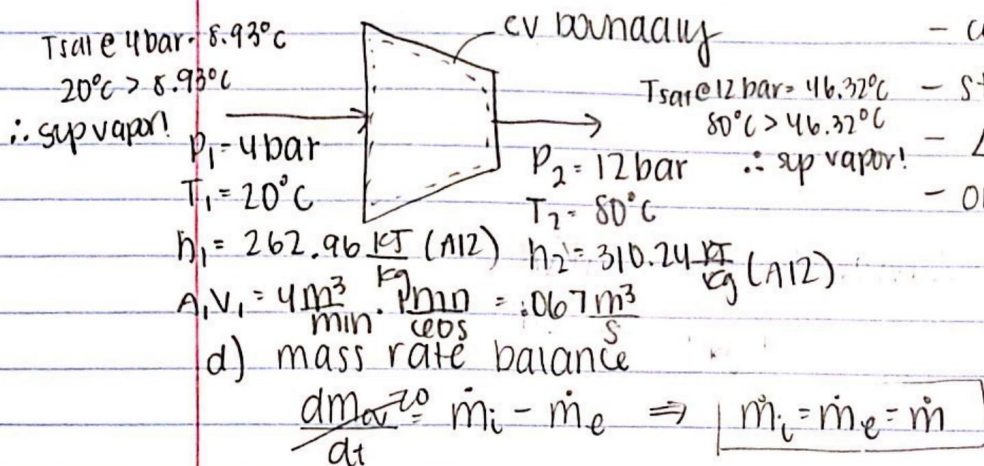
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- 0 pts Correct
- 1 pts a) Wrong schematic
- 1 pts a) Missing or wrong system boundary
- 2 pts b) State 1 is water vapor
- 4 pts b) Wrong or missing P-v diagram
- ✓ - 2 pts b) $V_2 > V_1$ (expansion)
- 2 pts b) State 2 is saturated vapor
- 2 pts c) Missing eng model
- 3 pts d) Missing simplified mass rate and energy rate balance.
- 1.5 pts d) Missing simplified mass rate
- 4 pts e) $W = 3581 \text{ kW}$

4.

Refrigerant 134a enters air conditioner compressor at 4 bar, 20°C, and is compressed at steady state to 12 bar, 80°C. The volumetric flow rate of R134a entering is 4 m³/min. The work input to the compressor is 60 kJ per kg of refrigerant flowing. Neglecting KE & PE, determine $\dot{Q}_{cv}$ (kW)

schematic:



engineering model

- control volume
- steady state
- $\Delta KE = \Delta PE = 0$
- one inlet / one outlet

energy rate balance

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) + g(z_1 - z_2) \right]$$

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}(h_1 - h_2)$$

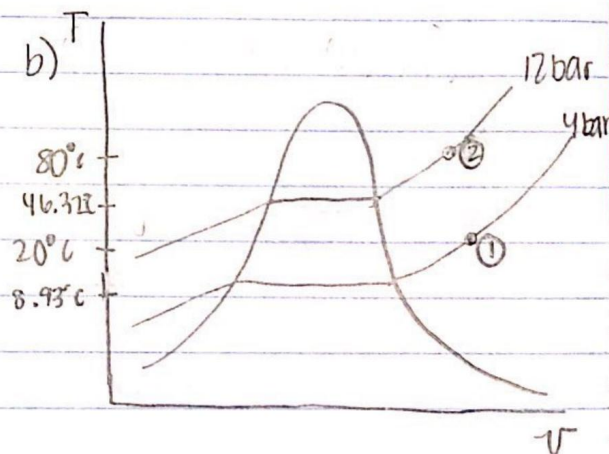
$$\dot{m} = \frac{A_1 V_1}{V_1} \rightarrow 0.05397 \frac{\text{m}^3}{\text{kg}} \text{ (A12)}$$

$$\dot{m} = \frac{0.067 \frac{\text{m}^3}{\text{s}}}{0.05397 \frac{\text{m}^3}{\text{kg}}} = 1.24 \frac{\text{kg}}{\text{s}}$$

(kJ/s) $\dot{W}_{cv} = \overset{\text{negative bc input}}{-60 \frac{\text{kJ}}{\text{kg}}} \cdot 1.24 \frac{\text{kg}}{\text{s}} = -74.12 \frac{\text{kJ}}{\text{s}}$

$$0 = \dot{Q}_{cv} + (-74.12 \frac{\text{kJ}}{\text{s}}) + 1.24 \frac{\text{kg}}{\text{s}} (262.96 \frac{\text{kJ}}{\text{kg}} - 310.24 \frac{\text{kJ}}{\text{kg}})$$

$$0 = \dot{Q}_{cv} + 15.71 \Rightarrow \boxed{\dot{Q}_{cv} = -15.71 \frac{\text{kJ}}{\text{s}}}$$



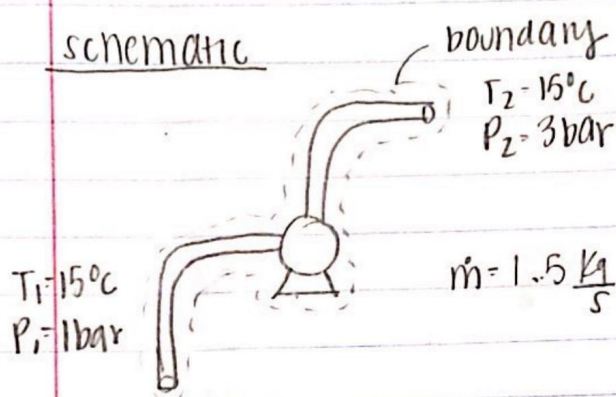
4 4 20 / 20

✓ - 0 pts Correct

- 3 pts T-V figure
- 1 pts T-V figure, $v_1 = ?$ $v_2 = ?$
- 5 pts Q calculation incorrect
- 3 pts scheme incorrect, you mention $\Delta PE = 0$ but your input and output is not on the same level
- 1 pts T-V figure where is step 1 and 2
- 3 pts scheme incorrect. input? output?
- 5 pts m incorrect
- 5 pts scheme

5.

A pump at steady state is drawing water @ $T_1 = 15^\circ\text{C}$ & $P_1 = 1\text{ bar}$ from a reservoir & delivering it at $P_2 = 3\text{ bar}$ to storage tank that is 15m above reservoir. The mass flow rate of water is 1.5 kg/s . T remains nearly constant @ 15°C & $\Delta KE = 0$, $\dot{Q}_{cv} = 0$. Determine power req by pump (kW)



engineering model

- control volume
- steady state
- one inlet / one outlet
- constant T
- ΔKE & $\dot{Q}_{cv} = 0$
- liquid water $v = v_f$
- $g = 9.81 \text{ m/s}^2$

c) mass rate balance

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e \Rightarrow \boxed{\dot{m}_i = \dot{m}_e = \dot{m}}$$

energy rate balance

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \frac{(v_1^2 - v_2^2)}{2} + g(z_1 - z_2) \right]$$

$$\dot{W}_{cv} = \dot{m} [(h_1 - h_2) + g(z_1 - z_2)]$$

$$h_1 - h_2 \Rightarrow [h_f(T_1) + v_f(T_1)[P_1 - P_{sat}(T_1)]] - [h_f(T_2) + v_f(T_2)[P_2 - P_{sat}(T_2)]]$$

constant T

$$\left(\frac{\text{kJ}}{\text{kg}} \right) h_1 - h_2 = v_f(T) [P_1 - P_2] = 1.0009 \times 10^{-3} \frac{\text{m}^3}{\text{kg}} (1\text{ bar} - 3\text{ bar}) \cdot \frac{10^5 \text{ N/m}^2}{1\text{ bar}} \cdot \frac{\text{kg}}{10^3 \text{ Nm}} = -0.20018 \frac{\text{kJ}}{\text{kg}}$$

$$h_1 - h_2 = -0.20018 \frac{\text{kJ}}{\text{kg}}$$

$$g(z_1 - z_2) = 9.81 \frac{\text{m}}{\text{s}^2} (0\text{ m} - 15\text{ m}) \cdot \frac{1\text{ N}}{1\text{ kg m/s}^2} \cdot \frac{1\text{ kJ}}{10^3 \text{ Nm}} = -0.14715 \frac{\text{kJ}}{\text{kg}}$$

$$\dot{W}_{cv} = 1.5 \frac{\text{kg}}{\text{s}} \left(-0.20018 \frac{\text{kJ}}{\text{kg}} + (-0.14715 \frac{\text{kJ}}{\text{kg}}) \right) = \boxed{-0.52 \frac{\text{kJ}}{\text{s}}}$$

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- 0 pts Correct

✓ - 3 pts scheme, storage tank that is 15 m above
reservoir

- 3 pts storage tank that is 15 m above

reservoir; the pump and the storage tank should be on the same level

- 5 pts w calculation incorrect

- 3 pts unit in KW

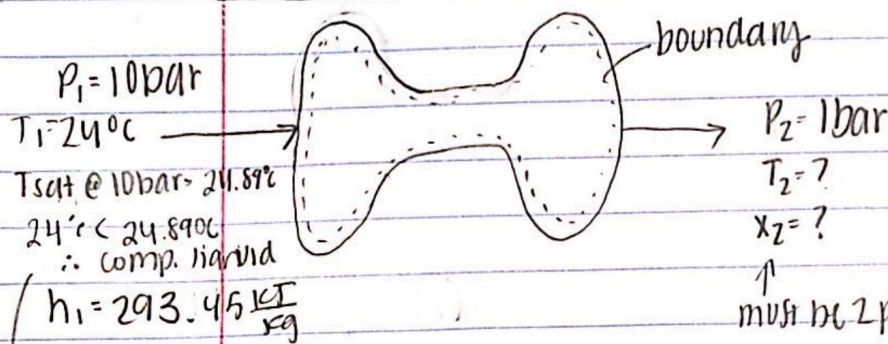
- 3 pts $h_1 = ?$ $h_2 = ?$

- 3 pts engineering model

b.

Ammonia enters the expansion valve of a refrigeration system at a $P_1 = 10 \text{ bar}$, $T_1 = 24^\circ\text{C}$. Exits at $P_2 = 1 \text{ bar}$. If refrigerant goes through throttling process, determine the quantity of refrigerant exiting the expansion valve.

schematic



engineering model

- steady state
- control volume
- one inlet / one outlet
- throttling process

$h_1 \approx h_f(24^\circ\text{C})$ c) mass rate balance

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e \Rightarrow \boxed{\dot{m}_i = \dot{m}_e = \dot{m}}$$

energy rate balance

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right) + g(z_1 - z_2) \right]$$

$$0 = \dot{m}(h_1 - h_2) \Rightarrow 0 = h_1 - h_2 \Rightarrow h_1 = h_2$$

$$h_1 = 293.45 \frac{\text{kJ}}{\text{kg}}$$

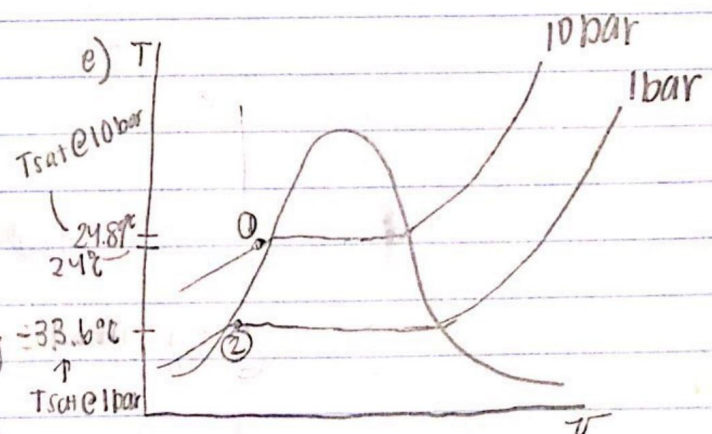
$$h_2 = h_{2f} + x_2 (h_{2g} - h_{2f})$$

$$h_{2f} = 28.18 \frac{\text{kJ}}{\text{kg}}$$

$$h_{2g} = 1398.41 \frac{\text{kJ}}{\text{kg}}$$

$$293.45 \frac{\text{kJ}}{\text{kg}} = 28.18 \frac{\text{kJ}}{\text{kg}} + x_2 \left(1398.41 \frac{\text{kJ}}{\text{kg}} - 28.18 \frac{\text{kJ}}{\text{kg}} \right)$$

$$\boxed{x_2 = 0.194}$$



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- 0 pts Correct
- 1 pts a) did not label the system boundary
- 1 pts a) wrong schematic
- 4 pts b) missing the engineering model
- 2 pts c) missing the mass rate balance
- 2 pts d) missing or wrong energy rate balance
- 4 pts d) used wrong enthalpy values to solve for x
- 5 pts d) $x = 0.194$
- 2.5 pts e) State 1 is compressed liquid
- 5 pts e) wrong or missing sketch of the process on a T-v diagram
- ✓ - 2.5 pts e) $v_2 > v_1$
- 2.5 pts e) missing point 2 on the T-v diagram
- 2.5 pts e) $T_1 > T_2$